Danny Muir

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EDUCATION

University of Washington, Seattle *Expected June 2027*
Bachelor of Science in Computer Engineering — **GPA: 3.84** Dean's List All Quarters 2023–2026
Paul G. Allen School of Computer Science & Engineering | **Minor:** Neural Computation and Engineering
Honors: Tau Beta Pi Engineering Honor Society; CSE Honors

TECHNICAL SKILLS

Programming: Python, Java, C, JavaScript, Bash, Git, HTML, LaTeX | **Platforms:** macOS, Linux, Windows
ML/AI: PyTorch, NumPy, Pandas, scikit-learn, MAML/Meta-Learning, Test-Time Adaptation, Autoencoders, Neural Networks, Computer Vision, Transformers
Neuroscience: ECoG/cortical probe signal processing, sEMG, neural decoding, temporal basis function models, closed-loop BCIs
Research: Grant writing, experimental design, project management, scientific computing, technical documentation

PUBLICATIONS

M. J. Bryan, **D. C. Muir**, F. Schwock, A. Yazdan-Shahmorad, R. P. N. Rao, "Cross-Session Pretraining Enables Robust Closed-Loop Neural Stimulation Through Meta-learning," *IEEE Transactions on Biomedical Engineering* (in preparation).

RESEARCH EXPERIENCE

Undergraduate Researcher Oct 2025–Present
Neural Systems Laboratory, UW
Advisor: Dr. Rajesh Rao, Paul G. Allen School of Computer Science & Engineering

- Optimizing multi-session neural decoding framework in collaboration with Matthew Bryan, building on temporal basis function models for closed-loop neural stimulation (Bryan et al., arXiv:2507.15274).
- Testing and selecting optimal implementations for session-specific embeddings and test-time adaptation methods.
- Improving training efficiency, model complexity, and cross-session generalization through systematic empirical evaluation.
- Conducting honors thesis research on learning dynamical systems underlying traveling waves in motor cortex to forecast motor-relevant neural subspaces.
- Beginning summer project integrating Clone-Structured Cognitive Graphs into the Active Predictive Coding framework to model cognitive map formation via predictive coding and hierarchical RL.

Associate Researcher, VP Operations Jan 2025–Oct 2025
Sensors, Energy, and Automation Laboratory (SEAL), UW
Advisor: Dr. Alexander Mamishev, Department of Electrical & Computer Engineering

- Conducted VLF/ULF signal processing research for atmospheric and geophysical monitoring applications.
- Contributed written sections and managed submission logistics for federal and industry grant proposals (DOE STTR, NIH, NSF, CDC, Google).
- Led project management team developing custom automation and DevOps solutions for laboratory operations.

GRANTS AND FUNDING

Federal Research Proposals (2025): Contributed written sections and managed submission logistics for DOE STTR, NIH, NSF, and CDC proposals on automation, sensor technology, and biomedical applications

Industry Research Proposals (2025): Contributing researcher on Google and industry partner applications for ML and automation research

LEADERSHIP AND SERVICE

President Nov 2025–Present

Triangle Fraternity, Washington Chapter

- Established Triangle UW's first Interfraternity Council (IFC) membership, expanding chapter participation in campus Greek life programming, philanthropy infrastructure, and inter-chapter relationships.
- Scaled sorority collaborations from ∼1 event per quarter to 11 events across two quarters (4 partner chapters, 2 new); chapter grew from 21 to 40 members with near-complete retention.
- Rebuilt executive board from 6 to 8+ formalized roles — adding Housing and Communications Directors and formalizing Academic Chair and Rush Chairs — distributing operational ownership and building institutional continuity.
- Chapter GPA above all-IFC and all-men's benchmarks every quarter; 66% of members on Dean's List; 100% new member attendance across all IFC conduct programs; 125+ combined service hours for the Leukemia & Lymphoma Society.

Vice President & New Member Educator Dec 2023–Nov 2025

Triangle Fraternity, Washington Chapter

- Built and systematized chapter event operations, enabling consistent execution of social and brotherhood programming.
- Served as chapter academic lead; chapter earned the Kahlert Academic Excellence Award (highest GPA of any Triangle chapter nationally) and Most Improved Chapter recognition from national Triangle.
- Redesigned new member education program, improving integration processes and developmental outcomes.

RESEARCH FOCUS & CAREER OBJECTIVE

Specializing in machine learning applications for neurotechnology and biomedical engineering, neuromorphic computing, and neural computation and engineering. Seeking research opportunities to advance neurotechnology and neural computation through interdisciplinary collaboration and ML-driven development.

References available upon request